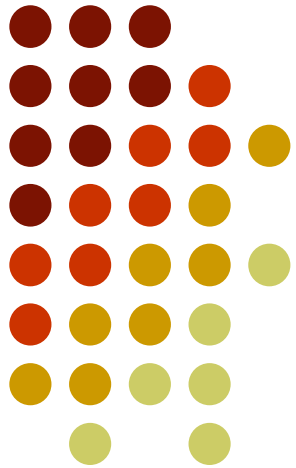
10 CCAPM及其讨论



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概要

- 基于消费的实利率理论
- 风险溢价及其测度
- CCAPM与真实世界：两个谜题

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10.1基于消费的实利率理论

- 回忆

$$1 = E\left[\tilde{\pi}\left(1+\tilde{r}_j\right)\right]$$

$$1 = E\left[\tilde{\pi}\left(1+r_f\right)\right]$$

$$r_f = \frac{1}{E\left(\tilde{\pi}\right)} - 1$$

$$\tilde{\pi} \triangleq \rho \frac{u'\left(\tilde{C}_1\right)}{u'\left(C_0\right)}$$

10.1基于消费的实利率理论

- 定义消费增长率 $\tilde{g} \triangleq \frac{\tilde{C}_1}{C_0} - 1$

$$\tilde{\pi} \triangleq \rho \frac{u'\left(\tilde{C}_1\right)}{u'\left(C_0\right)} = \rho \frac{u'\left[C_0\left(1+\tilde{g}\right)\right]}{u'\left(C_0\right)}$$

$$\approx \frac{\rho}{u'\left(C_0\right)}\left[u'\left(C_0\right) + u''\left(C_0\right)C_0\tilde{g} + \frac{1}{2}u'''\left(C_0\right)C_0^2\tilde{g}^2\right]$$

$$= \rho\left\{1 - \left[-\frac{C_0u''\left(C_0\right)}{u'\left(C_0\right)}\right]\tilde{g} + \frac{1}{2}\left[-\frac{C_0u''\left(C_0\right)}{u'\left(C_0\right)}\right]\left[-\frac{C_0u'''\left(C_0\right)}{u''\left(C_0\right)}\right]\tilde{g}^2\right\}$$

$$= \rho\left(1 - R_R\tilde{g} + \frac{1}{2}R_RP_R\tilde{g}^2\right)$$

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10. 1基于消费的实利率理论

$$E(\tilde{\pi}) = E\left[\rho\left(1 - R_R\tilde{g} + \frac{1}{2}R_R P_R \tilde{g}^2\right)\right]$$

$$= \rho\left[1 - R_R E(\tilde{g}) + \frac{1}{2}R_R P_R E(\tilde{g}^2)\right]$$

$$\approx \rho\left(1 - R_R\bar{g} + \frac{1}{2}R_R P_R \sigma_{\tilde{g}}^2\right)$$

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10.1基于消费的实利率理论

$$r_f = \frac{1}{E(\tilde{\pi})} - 1$$

$$\approx \frac{1}{\rho\left(1 - R_R\bar{g} + \frac{1}{2}R_R P_R \sigma_{\tilde{g}}^2\right)} - 1$$

$$\approx \frac{1-\rho}{\rho} + R_R\bar{g} - \frac{1}{2}R_R P_R \sigma_{\tilde{g}}^2$$

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10.1基于消费的实利率理论

- 真实无风险利率的决定因素
  - 参与者的“不耐”程度

$$\delta \triangleq \frac{1}{\rho} - 1$$

  - 经济增长和相对风险厌恶系数

$$R_R \bar{g}$$

  - 预防性储蓄动机

$$\frac{1}{2} R_R P_R \sigma_{\tilde{g}}^2$$

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10. 2风险测度及其溢价

$$E\left(\tilde{r}_j\right)-r_f=-\left(1+r_f\right)Cov\left(\tilde{\pi},\tilde{r}_j\right)$$

$$E\left(\tilde{r}_j\right)-r_f=-\left(1+r_f\right)Cov\left[\rho\frac{u'\left(\tilde{C}_1\right)}{u'\left(C_0\right)},\tilde{r}_j\right]$$

$$=-\frac{\rho\left(1+r_f\right)}{u'\left(C_0\right)}Cov\left[u'\left(\tilde{C}_1\right),\tilde{r}_j\right]$$

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10. 3两个谜题

- 无风险利率之谜
- 假定参与者的效用函数为CRRA型

$$u(w)=\frac{1}{1-\gamma}w^{1-\gamma},\gamma>0且\gamma\neq1.$$

- 有 $A(w)=\frac{\gamma}{w},R(w)=\gamma,P_R=\gamma+1$
- 故 $r_f=\delta+\gamma\bar{g}-\frac{1}{2}\gamma(\gamma+1)\sigma_{\tilde{g}}^2$

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10. 3两个谜题

- 考虑一个经济
  - 时间贴现因子ρ为0.98，相对风险厌恶系数为2，总消费增长率1.8%，增长率的波动3.6%

$$r_f = \delta + \gamma \bar{g} - \frac{1}{2}\gamma(\gamma+1)\sigma_{\tilde{g}}^2$$

$$= 2.04\% + 2 \times 1.8\% - \frac{1}{2} \times 2 \times 3 \times 3.6\%^2$$

$$= 5.25\%$$

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10. 3两个谜题

- 中国经济统计数据
  - 2000-2019年间，经济增长率平均为9.01%，增长率的波动为2.13%
  - 真实无风险利率

$$r_f = \delta + \gamma \bar{g} - \frac{1}{2}\gamma(\gamma+1)\sigma_{\tilde{g}}^2$$

$$= 2.04\% + 2 \times 9.01\% - \frac{1}{2} \times 2 \times 3 \times 2.13\%^2$$

$$= 19.92\%$$

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10. 3两个谜题

- 风险溢价之谜
- 仍假定参与者的效用函数为CRRA型

$$u(w)=\frac{1}{1-\gamma}w^{1-\gamma},\gamma>0且\gamma\neq 1.$$

- 随机贴现因子为

$$\tilde{\pi}=\rho\frac{u'(\tilde{C}_1)}{u'(C_0)}=\rho\frac{\tilde{C}_1^{-\gamma}}{C_0^{-\gamma}}=\rho(1+\tilde{g})^{-\gamma}\approx\rho(1-\gamma\tilde{g})$$

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10. 3两个谜题

- 风险溢价

$$
\begin{aligned}
E\left(\tilde{r}_j\right)-r_f &= -\left(1+r_f\right)Cov\left(\tilde{\pi},\tilde{r}_j\right) \\
&= -\left(1+r_f\right)Cov\left[\rho\left(1-\gamma\tilde{g}\right),\tilde{r}_j\right] \\
&= \rho\gamma\left(1+r_f\right)Cov\left(\tilde{g},\tilde{r}_j\right) \\
&\approx \rho\gamma\left(\frac{1}{\rho}+\gamma\bar{g}\right)Cov\left(\tilde{g},\tilde{r}_j\right) \\
&= \gamma\left(1+\rho\gamma\bar{g}\right)Cov\left(\tilde{g},\tilde{r}_j\right)
\end{aligned}
$$

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10. 3两个谜题

- Mehra & Prescott (1985)
  - 1889-1978年间美国经济统计数据

无风险利率 r_f	0.8%
股票平均收益 $E\left(r_{S\&P500}\right)$	6.98%
股票收益率的标准差 $\sigma_{S\&P500}$	16.5%
消费的平均增长 $\overline{g}$	1.8%
消费增长率的标准差 $\sigma_{\tilde{g}}$	3.6%
股票收益率和消费增长率之间的相关系数	0.5

10. 3两个谜题

$$6.18\% = \gamma\left(1+\gamma\times 0.98\times 1.8\%\right)\times 0.3\%$$

$$\gamma = 20.68$$

- γ高得离谱！
- 如果接受如此之高的γ，那么真实无风险利率

$$r_f = \delta + \gamma\bar{g} - \frac{1}{2}\gamma\left(\gamma+1\right)\sigma_{\tilde{g}}^2$$

$$= 2.04\% + 20.68\times 1.8\% - \frac{1}{2}\times 20.68\times 21.68\times 3.6\%^2$$

$$= 10.21\%$$

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10. 3两个谜题

- 对两个谜题的进一步探索
  - 修改效用函数
    - 广义预期效用函数， Epstein-Zin(1989, 1991)
    - 消费惯性，放弃时间可分离性
  - 不完全市场
  - 市场摩擦
  - 行为金融

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